

BVAR: Bayesian Vector Autoregressions with Hierarchical Prior Selection in R

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The Paper

Motivation: The Curse of Dimensionality

Vector Autoregression (VAR) jointly models M time series over p lags:

$$\mathbf{y}_t = \mathbf{a}_0 + A_1 \mathbf{y}_{t-1} + \cdots + A_p \mathbf{y}_{t-p} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \Sigma)$$

The problem:

- Parameter count: $M + M^2 p \rightarrow$ grows quadratically
- $M=6$, $p=5$: already **186 parameters**
- OLS overfits; poor out-of-sample forecasting

Bayesian solution:

- Encode prior knowledge via $p(\boldsymbol{\theta})$
- Regularise via **shrinkage priors**
- Posterior $\propto$ Likelihood $\times$ Prior

Key challenge

How to choose the prior's hyperparameters in a principled, data-driven way?

Intuition: Why Does Dimensionality Matter?

The core problem:

- More variables and lags $\Rightarrow$ exponentially more parameters
- OLS must estimate all of them from limited data
- The model fits noise rather than signal

Why Bayesian shrinkage helps:

- A prior acts as a soft constraint: it shrinks estimates toward a simple, persistent benchmark
- Lets the data override the prior only where evidence is strong
- Better out-of-sample forecasts with far fewer “effective” parameters

Key takeaway: Without regularisation, a VAR with many variables is unreliable. Bayesian priors are a principled way to regularise.

The Minnesota Prior

Core idea (Litterman 1980): each variable is a priori a random walk.

$$\mathbb{E}[(A_s)_{ij} \mid \Sigma] = \begin{cases} 1 & i = j, s = 1 \text{ (first lag)} \\ 0 & \text{otherwise} \end{cases}$$

$$\text{Var}[(A_s)_{ij} \mid \Sigma] \propto \frac{\lambda^2 \psi_j}{s^{2\alpha}} \cdot \frac{\sigma_{ii}}{\sigma_{jj}}$$

Key hyperparameters:

- λ - overall tightness
($\lambda \rightarrow 0$: only prior; $\lambda \rightarrow \infty$: only data)
- α - lag-decay speed
- ψ_j - per-variable scaling

Refinements:

- **Sum-of-coefficients** (μ): shrinks coefficients so persistent variables behave like they stay near their current level
- **Single-unit-root** (δ): shrinks the system toward common stochastic trends, helping accommodate cointegration

Open problem: setting λ, μ, δ manually is ad hoc and dataset-specific.

Intuition: Why a Random Walk Prior?

The economic logic:

- Most macro and financial series are persistent - today's value is a good predictor of tomorrow's
- Random walk encodes this: "absent other information, things stay roughly the same"
- Cross-variable effects are shrunk toward zero - a variable rarely drives another strongly across lags

Key takeaway: The random walk prior is not an assumption we believe literally – it is a useful regularisation device grounded in stylised facts about time series.

Hierarchical Prior Selection (Giannone et al. 2015)

Key idea: treat hyperparameters $\gamma = (\lambda, \mu, \delta, \dots)$ as random variables.

Two-level Bayesian model:

$$p(\theta, \gamma \mid \mathbf{y}) \propto \underbrace{p(\mathbf{y} \mid \gamma)}_{\text{marginal likelihood}} p(\gamma)$$

$$p(\mathbf{y} \mid \gamma) = \int p(\mathbf{y} \mid \theta, \gamma) p(\theta \mid \gamma) d\theta$$

- Marginal likelihood **penalises complexity** automatically (Occam's razor)
- With conjugate NIW prior $\Rightarrow$ **closed-form** $p(\mathbf{y} \mid \gamma)$
- γ drawn via **Metropolis-Hastings**; θ drawn analytically

Practical benefit

Optimal shrinkage is learned from data - no manual tuning required.

Intuition: Why Learn Hyperparameters from Data?

The problem with manual tuning:

- Different datasets need different amounts of shrinkage
- Short samples: tighter prior helps; long samples: data should dominate
- Manually choosing λ is arbitrary and hard to justify

The hierarchical solution:

- Marginal likelihood rewards priors that make the data probable
- It automatically penalises models that are too complex or too rigid
- Can be thought of as “Bayesian cross-validation” built into the estimation

Key takeaway: Instead of guessing the right amount of shrinkage, the model learns it. This makes results reproducible and principled across different datasets and sample sizes.

Intuition: IRF, FEVD and GFEVD

Impulse Response Function (IRF): traces how a one standard deviation shock to variable j propagates to variable i over time.

- Answers: *“If US volatility spikes today, by how much does European volatility rise in 5 days?”*
- Recursive (Cholesky) identification assumes a causal ordering; sign-restrictions are an alternative

Forecast Error Variance Decomposition (FEVD): breaks down the h -step forecast uncertainty of variable i into contributions from each structural shock.

- Answers: *“What fraction of Japan’s volatility forecast error is driven by US shocks?”*

Generalised FEVD (GFEVD) (Pesaran & Shin 1998): same idea, but using generalised impulses – **no ordering required**.

- Robust to variable ordering; especially useful when no clear causal hierarchy exists across markets

The BVAR R Package

Core functions:

1. `bv_minnesota()` - specify prior
2. `bvar(data, lags = p, ...)` - estimate
3. `irf()` / `predict()` - analyse

Conjugate Normal-Inverse-Wishart (NIW)

setup (multivariate generalization of the Normal-Inverse-Gamma prior):

$$\beta \mid \Sigma \sim \mathcal{N}(\mathbf{b}, \Sigma \otimes \Omega)$$

$$\Sigma \sim \mathcal{IW}(\Psi, d)$$

⇒ closed-form posteriors; MH only for γ .

Features:

- Recursive & sign-restricted IRFs
- Conditional forecasting (scenario analysis)
- FRED-MD / FRED-QD datasets integration
- coda interface for convergence diagnostics
- BVARverse for ggplot2 output

Our Project

Our Project: Data & Research Question

Research question: How do volatility shocks transmit across global markets, and does this transmission change during the COVID-19 crisis?

VAR and BVAR applications to macroeconomic and financial variables have proven to be useful tools for studying cross-market dynamics and shock transmission.

Data: daily implied volatility indices, 2015-2025 ($M = 6$, $p = 5$): **186 parameters**

Index	Market	Underlying
VIX	United States	S&P 500
V2TX	Euro Area	EURO STOXX 50
IVIUK	United Kingdom	FTSE 100
VNKY	Japan	Nikkei 225
VHSI	Hong Kong	Hang Seng
India VIX	India	Nifty 50

Sub-sample comparison:

- **Calm market:** 2015-2019
(low global volatility)
- **Crisis:** 2020-2021
(COVID-19 shock)

Our Project: Methodology & Planned Outputs

Model: Hierarchical BVAR

- Minnesota prior with auto-tuned λ (Giannone et al. 2015)
- Implemented via BVAR R package
- Estimated separately on calm and crisis sub-samples

Planned outputs:

1. **IRFs** - 1 s.d. US VIX shock; response of all markets over 30-day horizon
2. **GFEVD** - share of each market's forecast variance explained by US shocks
3. **Sub-sample comparison** - do IRFs & GFEVDs differ significantly between calm and COVID-19 crisis periods?

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